

Reading Response Evaluation Sheet (70%)

No.	Aspect	Marks
1.	Text Engagement	/ 20
2.	Personal Reflection	/ 20
3.	Critical Thinking	/ 10
4.	Language	/ 10
5.	Organization	/ 10
TOTAL		/ 70

Note: Please place this after the cover page

Tutor's Signature: _____

Date: _____

As a firm believer that the finest approach to learning is to engage all our senses, virtual reality strikes me as the perfect medium to deliver education. This article discusses the impact of virtual reality—the opportunities, drawbacks, and prospects—in healthcare education. The author also highlights the various approaches—screen-based learning, 360-video, and interactive virtual reality—to delivering education to young medical students. While acknowledging the expenses associated with maintaining a virtual reality learning environment, he also emphasises how virtual reality may outperform conventional tools used in delivering an effective learning experience. This reading response will focus on the advantages, prospects, and the caveats of virtual reality in healthcare education.

The author states that "**the psychological safety, enjoyable nature, and potential for gamification of VR encourage engagement and autonomous learning.**" He had implied this notion earlier in the article by drawing a comparison between virtual reality and a hyperrealistic computer game. Similar to video games, the participants are aware that they are in virtual settings; this awareness boosts their confidence, as they know any mistakes made present no real harm. Emboldened by this, they are more motivated to apply their knowledge in the simulation. Furthermore, I believe this awareness reduces their stress, allowing them to conduct each procedure calmly and inadvertently ingrain the procedures in their minds smoothly. Subsequently, this will enable them to be adept at handling specific scenarios before working with actual, erratic patients, making them to be less prone to panicking. Nonetheless, my nuance is that some learners may treat the simulations casually, as they believe the simulation does not accurately depict real-life scenarios, leading them to perform procedures haphazardly.

Although no virtual reality environment is within my reach, I find the same comfort when utilising a sandbox environment, as it allows me to experiment with my queries and methods before deploying them in my assignments or any major project. By creating a closed environment, I can apply the theories learnt from lectures to assess whether I have a firm grasp of the necessary knowledge of the topic before applying it to real-world tasks. A well-known sandbox environment that I utilise is LeetCode to experiment with various methods of problemsolving to questions similar to my assignments or tutorials; its sandbox characteristics frees me from having to worry about affecting any real-world software and hardware while experimenting with different algorithms and data structures. Just like the simulation, the problems in LeetCode may not represent reality well, however, I can still hone my foundational skills to the utmost or add new knowledge into my arsenal that will be used to solve real-life

problems. This idea may interest healthcare educators who are postponing the incorporation of virtual reality into their co-curriculum because they underestimate or disregard its positive impacts on students.

The author believes that eventually, **instead of simulations being a once-a-day, faculty-led activity, learners can engage in them whenever they wish.** Perhaps, his testament comes from the evident advancements of virtual reality; it has evolved from cumbersome, wired systems with external sensors to a streamlined, wireless experience with built-in tracking functions, easing the accessibility to the virtual environment. This can be observed through innovations such as Apple Vision Pro, Meta Quest headsets, and Sony PlayStation headsets that no longer require convoluted wiring but embrace plug-and-play technology. Furthermore, there are a plethora of headsets offered in the market that are offered at a lower cost with little compromise, making virtual reality headsets much more accessible than ever before.

As I process this notion, I cannot help but wonder, if this technology is available to the public, what's stopping medical schools and universities from making greater investments in virtual reality? He touches on this point briefly; he claims estimating the expenses is complicated as the costs of maintaining physical simulation are distinct between institutions. As I read his elaboration, I realised that institutions must cater to the masses rather than a select few, so they must conduct routine maintenance, and hardware issues can quickly rack up maintenance costs. Furthermore, a university must prioritise providing the optimal experience for every student. An individual at home can choose to disregard an update or cancel a subscription, but a university must keep its technology up to date to remain relevant.

Furthermore, I believe this point will elicit enthusiasm from the public, especially students. Humanity's rapid advancements in AI will soon allow virtual reality environments to be household names in educational environments across the globe. I find this prospect to be exciting as it will enable me to access clinical experiences that were previously unavailable. For instance, if given the chance, I can immerse myself in the hustle and bustle during an emergency rather than relying on TV shows to portray an inaccurate picture that represents only a slither of reality. Furthermore, adolescents interested in a healthcare career can safely observe and engage in various clinical settings, gaining a more realistic understanding of the responsibilities that healthcare personnel shoulder.

The author **questions the suitability of virtual reality in coaching students through several distinct scenarios**. He lists a few examples to clinch his point, each sharing a common theme – they are either too sophisticated or social interactions. He has a firm stance on virtual reality being a tool and not a panacea; he argues that the current artificial intelligence technology is still immature to cover the more crucial elements of a clinical setting. I concur with his belief that facial expressions, tones, slang, and language nuances are most effectively conveyed by a human. After all, virtual characters and scenarios are merely programmed scripts that lack independent thoughts. Despite countless AI dystopian depictions have led us to believe, our current technology cannot even graze the far-fetch capabilities presented in movies let alone create a convincing artificial human being. For instance, video games often struggle to implement facial expressions comparable to that of a real person due to hardware and software limitations. Every video game I played did not feature characters whose mouths or facial expressions were synchronised with their dialogs. Simulations are not exempted from this limitation; a simulated doctor may speak with tones and nuances, but their facial expressions do not match, attenuating the immersive experience.

I firmly believe that any social interactions – breaking bad news, presenting a diagnosis, and discussing personal matters - with patients are best trained with a human as it is impossible to programme human emotions and cognitive functions - empathy and intuition - with our existing technology. In my opinion, training with humans will yield a genuine interaction filled with subtexts, tones, and expressions rather than robotic, predetermined responses from virtual characters. Despite being involved in the technology industry, ironically, I find his argument rather consoling. Currently, an increasing number of companies is adopting the AI model, forcing me to interact with an AI rather than a human. I am not denying its convenience, I am denying its replacement of humanity. A prominent example is the implementation of chatbots which, quite frankly, is frustrating to utilise. Often, the chatbots are subpar to the likes of OpenAI or Google AI by a wide margin, especially in their ability to read context-specific or nuanced queries. Furthermore, simulated characters lack the intuition and nuanced decisionmaking skills that humans possess. Nonetheless, with so many figures participating in the AI race, it is only a matter of time before an artificial human is invented. This may be an interesting point of exploration to educators who are designing the programme structure as they need to categorise activities that can be coached via virtual reality and those that cannot. Understanding the potential and limitations of virtual reality can assist them to create a balanced

curriculum that exploits the immersive capabilities virtual reality offers while reserving human-led sessions for areas requiring social and emotional intelligence.

All in all, I believe this article would be an interesting read for those who are looking for surface-level information regarding integrating virtual reality in healthcare education, specifically medical students, lecturers, and education institutions. This article is effective in providing general information about the positives, negatives, and prospects of this technology. The author does not provide statistics on specific observed benefits and drawbacks amongst students to prove his statements. He does, however, highlight the universities that are using virtual reality along with its feedback. If a reader wishes to have a more in-depth view into the implementation of virtual reality, a research article would be more suitable. Thus, this article serves as an introduction into learning more about the implementation of virtual reality in healthcare education.

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